DS4I Assignment 2

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Siamese Neural Network

Methods

General Siamese structure

- Input: 2 images
- Uses a CNN
- Output: distance between embeddings of images
- Distance infers dissimilarity
- Contrastive loss:

$$L = yD^2 + (1-y)max(0, m-D)^2$$

Input pairs

Training:

Validation:

Test:

Input pairs

Training:

- sessions 1, 2, 3

Validation:

Test:

Input pairs

Training:

- sessions 1, 2, 3

Validation:

- session 4

Test:

Input pairs

Training:

- sessions 1, 2, 3

Validation:

- session 4

Test:

- same writer, same session

Input pairs

Training:

- sessions 1, 2, 3

Validation:

- session 4

Test:

- same writer, same session
- same writer, different sessions

Input pairs

Training:

- sessions 1, 2, 3

Validation:

- session 4

Test:

- same writer, same session
- same writer, different sessions
- different writers, same session

Input pairs

Training:

- sessions 1, 2, 3

Validation:

- session 4

Test:

- same writer, same session
- same writer, different sessions
- different writers, same session
- different writers, different sessions

Input pairs

Training:

- sessions 1, 2, 3

Validation:

- session 4

Test:

- same writer, same session
- same writer, different sessions
- different writers, same session
- different writers, different sessions
- sessions 1 - 5

Metrics

Sensitivity

- Proportion of actual positive cases correctly identified

$$\text{Sensitivity} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Specificity

- Proportion of actual negative cases correctly identified

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{True Negatives} + \text{False Positives}}$$

Balanced accuracy

- Accounts for imbalanced classes

$$\text{Balanced Accuracy} = \frac{\text{Sensitivity} + \text{Specificity}}{2}$$

Default calibration

Hyperparameter	Value
Hidden layers	2
Activation functions	ReLU
Number of filters	16 and 2×16
Filter size	3×3
Learning rate	1×10^{-3}
Dropout rate	0.2
L2 regularisation	1×10^{-3}
Layer 1 pooling window	2×2
Embedding length	16
Epochs	15

Tuning

- Hidden layers
- Activation functions
- Filter (square) size
- Number of filters
- **Average validation** performance metrics reported

Results

Tuning hidden layers

Number of Hidden Layers	Contrastive Loss	Balanced Accuracy
1	0.214	0.773
2	0.105	0.966

Tuning activation functions

Activation Functions	Contrastive Loss	Balanced Accuracy
relu_relu	0.105	0.966
tanh_relu	0.103	0.971
relu_tanh	0.108	0.982
tanh_tanh	0.113	0.980

Tuning filter numbers and size

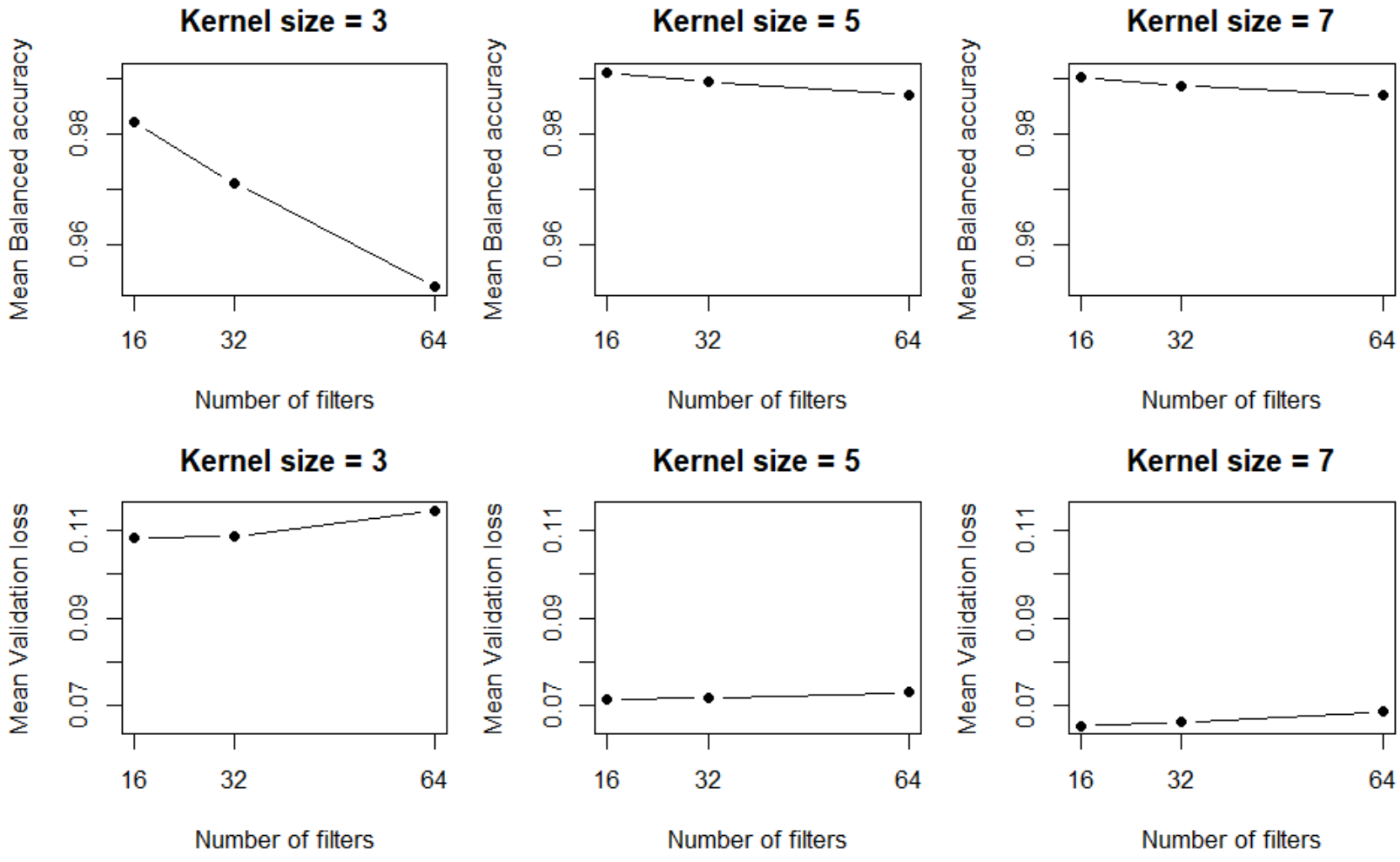

Figure 1

Final model

Tuned inputs

- Hidden layers: 2
- Activation functions: reLU and tanh
- Filter (square) size: 5×5
- Number of filters: 16 and 32
- Epochs: 6

Outputs

- Test Balanced accuracy: 0.751
- Test Loss: 0.104
- Distance threshold: 0.247

Session results

Session Type	n	Sensitivity	Specificity	Balanced Accuracy
Same	2000	0.899	0.995	0.947
Different	2000	0.112	0.997	0.554

Digit results

Digit	Sensitivity	Specificity	Balanced Accuracy
0	0.501	0.995	0.748
1	0.532	0.995	0.763
2	0.513	0.998	0.755
3	0.476	1.000	0.738
4	0.510	0.992	0.751

Writer results (1 - 10)

Writer	Sensitivity	Specificity	Balanced Accuracy
1	0.538	0.991	0.765
2	0.432	1.000	0.716
3	0.481	1.000	0.740
4	0.518	1.000	0.759
5	0.476	0.988	0.732
6	0.444	0.990	0.717
7	0.587	1.000	0.793
8	0.429	0.991	0.710
9	0.523	1.000	0.761
10	0.557	1.000	0.778

Writer results (11 - 20)

Writer	Sensitivity	Specificity	Balanced Accuracy
11	0.495	1.000	0.748
12	0.520	1.000	0.760
13	0.403	0.991	0.697
14	0.453	0.983	0.718
15	0.484	1.000	0.742
16	0.543	1.000	0.772
17	0.542	0.987	0.765
18	0.613	1.000	0.806
19	0.557	1.000	0.778
20	0.552	1.000	0.776